

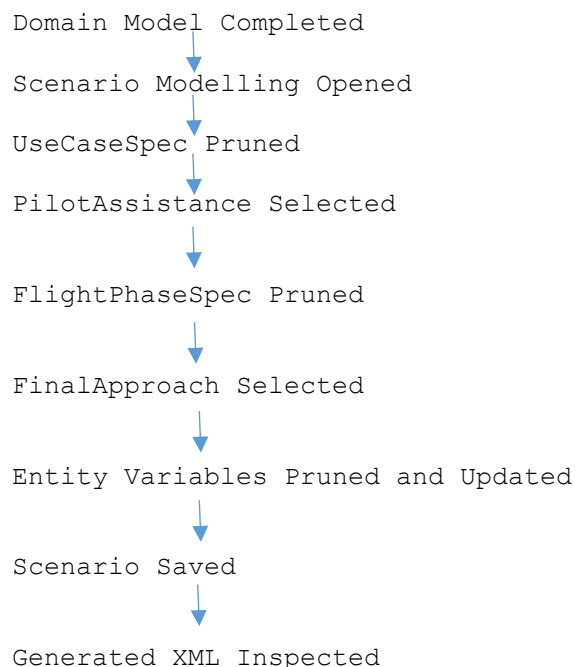
Scenario Modelling, Pruning, and Functional Verification

1. Objective

The purpose of this stage was to verify whether a user could derive a scenario-specific model from the Pilot Assistance domain model and whether ODME preserved the selected structure and parameter values.

2. Scenario Workflow

The following workflow was executed:



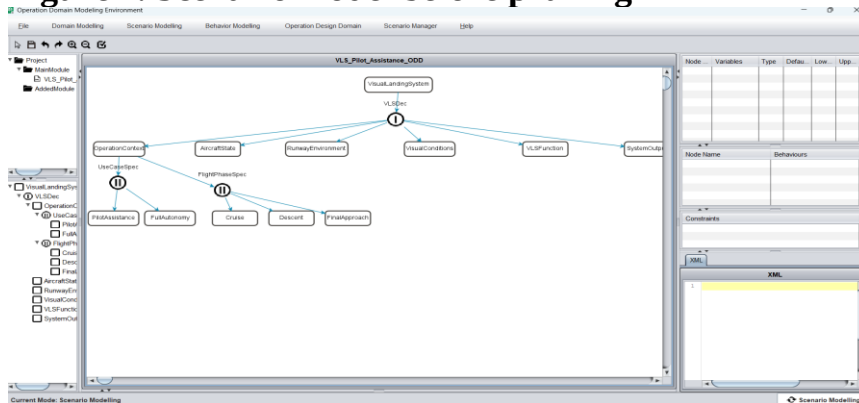
A later variation changed the flight phase selection from `FinalApproach` to `Cruise`. The generated XML subsequently contained:

```
Cruise_PilotAssistance_OperationContext
```

This was consistent with the intentional change in pruning selection.

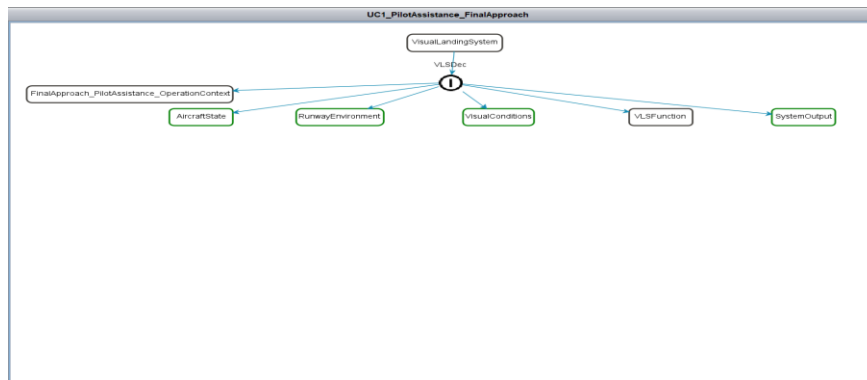
3. Structural Pruning Verification

Figure 1. Scenario model before pruning



Scenario Model before pruning

Figure 2. Scenario model after specialization pruning



Observation:

The scenario representation reflected the selected operational alternatives.

4. Variable Pruning Verification

Variables were updated through the pruning workflow for:

- AircraftState,
- RunwayEnvironment,
- VisualConditions,
- SystemOutput.

The values remained visible after switching between Domain Modelling and Scenario Modelling.

Result: Pass.

5. Scenario Persistence Verification

Expected result:

Updated scenario values should remain available after navigation between modelling modes and after scenario save.

Actual result:

The updated values remained available.

Status: Pass.

6. XML Generation Verification

After saving the scenario, ODME generated XML containing:

- the selected operational context
- AircraftState variables
- RunwayEnvironment variables
- VisualConditions variables
- VLSFunction behaviours
- SystemOutput variables.

The generated XML values matched the values selected during scenario modelling.

Status: Pass for the values checked.